

PORTER COUNTY AP, IN, USA

WMO: 725327

Lat: 41.453N

Lon: 87.006W

Elev: 235

StdP: 98.54

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 04846

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.4	-15.8	-22.5	0.5	-17.8	-19.9	0.7	-15.2	11.9	-2.2	10.9	-4.2	4.0	280	0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	32.5	23.5	31.2	22.9	29.8	22.0	25.4	30.3	24.4	28.9	23.5	27.9	4.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.8	19.2	28.5	22.8	18.1	27.0	22.2	17.4	26.3	78.8	30.3	74.7	29.0	71.2	27.9	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	9.1	8.2	DB	-22.4	34.9	3.4	2.0	-24.9	36.3	-26.9	37.5	-28.9	38.6	-31.4	40.1	(4)
(5)			WB	-22.7	27.1	3.4	1.1	-25.2	27.9	-27.1	28.5	-29.0	29.2	-31.5	30.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	-4.1	-2.6	3.5	9.9	15.8	21.0	23.0	21.9	18.5	11.9	5.1	-1.2	(6)	
	DBStd	10.79	6.52	6.19	6.02	5.23	5.04	3.80	3.18	2.83	4.18	5.04	5.26	6.08	(7)	
	HDD10.0	1631	436	355	218	65	8	0	0	0	1	37	162	350	(8)	
	HDD18.3	3394	694	587	462	256	112	17	3	4	48	210	396	606	(9)	
	CDD10.0	1742	1	2	16	63	188	331	403	369	257	95	16	2	(10)	
	CDD18.3	464	0	0	1	4	34	98	148	116	54	9	0	0	(11)	
	CDH23.3	4100	0	0	11	52	338	864	1343	918	491	83	1	0	(12)	
	CDH26.7	1321	0	0	2	7	99	292	473	271	159	17	0	0	(13)	
(14) Wind	WSAvg	3.7	4.6	4.3	4.2	4.5	3.8	3.2	2.7	2.5	2.8	3.6	4.1	4.3	(14)	
(15) Precipitation	PrecAvg	994	48	41	74	102	94	107	100	93	106	78	81	73	(15)	
	PrecMax	1229	106	115	142	205	171	245	288	258	262	174	214	167	(16)	
	PrecMin	644	7	5	14	13	32	18	26	12	1	13	35	16	(17)	
	PrecStd	149	27	25	32	46	39	50	53	57	63	39	42	39	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	17.1	24.8	27.6	31.9	33.7	34.8	32.9	33.0	28.9	21.4	16.2	(19)	
		MCWB	12.5	12.7	17.1	17.2	21.4	23.2	25.7	24.7	22.6	20.8	14.4	14.1	(20)	
	2%	DB	10.3	12.5	19.4	24.5	29.4	32.0	32.4	31.2	31.0	25.8	17.9	12.4	(21)	
		MCWB	8.2	10.1	14.0	15.9	19.7	22.4	24.4	23.5	22.1	18.2	12.9	10.3	(22)	
	5%	DB	7.3	8.4	16.2	22.3	27.3	30.0	31.0	29.6	28.0	22.8	15.7	9.3	(23)	
		MCWB	5.5	6.0	11.9	15.2	18.6	21.7	23.7	22.5	20.4	17.0	12.1	7.6	(24)	
	10%	DB	4.5	5.3	13.1	19.4	24.7	28.0	29.2	27.9	26.2	20.3	13.0	7.0	(25)	
		MCWB	3.0	3.2	9.5	13.1	17.9	20.8	22.5	22.0	19.4	15.9	9.9	5.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.1	14.0	17.5	20.0	23.3	25.5	27.0	26.2	24.5	22.1	16.2	14.3	(27)	
		MCDB	14.3	15.8	24.5	24.9	29.3	31.4	32.6	30.9	30.9	27.7	18.9	15.9	(28)	
	2%	WB	8.8	10.1	15.2	17.7	21.5	24.1	25.8	24.8	23.2	20.1	14.5	10.9	(29)	
		MCDB	9.7	12.5	18.5	22.4	26.9	29.2	30.7	29.3	28.2	23.4	17.1	12.0	(30)	
	5%	WB	5.5	6.5	12.7	15.8	20.2	23.0	24.8	23.9	21.9	18.2	12.4	7.7	(31)	
		MCDB	7.2	8.3	15.2	20.5	25.0	28.1	29.2	27.8	26.0	21.2	14.9	9.1	(32)	
	10%	WB	3.1	3.3	9.8	14.2	19.0	22.0	23.8	22.9	20.7	16.5	10.3	5.3	(33)	
		MCDB	4.4	5.2	12.9	19.1	23.5	26.5	28.0	26.6	24.6	20.5	13.1	6.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	8.1	9.6	11.4	11.3	10.9	10.8	10.6	11.7	10.3	8.6	7.3	(35)	
		MCDBR	10.0	11.6	14.2	15.6	14.0	12.7	11.9	12.4	13.9	13.6	11.9	10.0	(36)	
	5% WB	MCWBR	8.5	9.5	9.5	9.0	6.9	5.8	5.7	5.9	6.3	7.3	8.2	8.7	(37)	
		MCWBR	9.7	11.2	12.8	13.9	11.9	11.0	10.6	10.3	11.4	11.2	10.4	9.2	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.298	0.324	0.345	0.400	0.429	0.437	0.453	0.447	0.420	0.356	0.319	0.297	(40)	
		taud	2.425	2.343	2.380	2.241	2.216	2.236	2.187	2.200	2.244	2.428	2.483	2.473	(41)	
	Ebn at Noon	Ebn at Noon	856	880	903	870	850	841	823	816	811	834	824	827	(42)	
		Edh at Noon	78	99	108	133	140	138	143	137	123	91	74	70	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.60	2.45	3.58	4.63	5.27	6.05	6.08	5.30	4.40	2.90	1.83	1.34	(44)	
	RadStd	RadStd	0.16	0.32	0.28	0.39	0.47	0.38	0.31	0.31	0.36	0.30	0.22	0.14	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (56 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page